Is the leak-free network running DD?

Causal interventions, not probe scores

YES for DD's inference. NO for DD's search.

Recolour the ONE neighbour whose colour forces the answer -- a single token --
and the model follows DD's recomputed answer:

1.000	when that neighbour is a GIVEN hint	(round 0, n=350)
0.980	when it is one the model DEDUCED itself	(rounds 1-4)
0.388	leaked twin, the model finding 2 called 'not running DD'	

Chance is 0.25. The second line is the one that matters: it eliminates
against facts it produced, which is propagation, not a single rule application.

But the SCHEDULE is not DD's: events come out in ascending node index for
100% of rounds, and a no-deduction hand rule beats its own node features.

Jobs 50075339 (round 0) and 50077245 (chained), leak-free N=24, d=256 L=12 ff=1024, shuffle_adj=True. Chance = 1/n_colors = 0.25.

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING BEGIN_ROUND SOLVED UNRESOLVED	*section + round markers*
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3	*the q colours*
10-33	N0 N1 ... N23	*one id per node, no structure in the id*

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

The graph section is a neighbour table, not an image: node, then its degree neighbours. THE ORDER WITHIN A ROW IS THE LEAK of finding 16.

THIS INSTANCE, 198 TOKENS (seed 50000000004)

HINTS	BEGIN_HINTS N2 COLOR_2 N3 COLOR_2 N5 COLOR_2 N8 COLOR_3 N10 COLOR_0 N11 COLOR_1 ... (31 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N5 N18 N13 N14 N1 N8 N13 N2 N11 N19 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 N9 COLOR_1 N12 ... (22 tokens)

Loss is on the REASONING section only: 2 rounds, 9 deductions, 18 target tokens. Everything before BEGIN_REASONING is the problem statement.

One real reasoning trace

15 hints of 24 nodes, then 2 rounds of simultaneous deduction. Hints: N2=2 N3=2 N5=2 N8=3 N10=0 N11=1 N13=3 N14=0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 ...
  8 deduced
BEGIN_ROUND          N7 COLOR_2
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours [16, 5, 18, 13, 14]

colours known before this round:

```
[0, 2, 3]  (3 = q-1)
```

only colour left: 1

`so the trace emits  N0 COLOR_1`

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

The intervention edits ONE colour token in this section (chained arm) or in the hints (round-0 arm).

T=2 rounds, the median of 40 draws from this config (range 1-6); seed 50000000004, shuffle_adj=True, shuffle_events=False; evidence from sudoku_tokens.evidence_rounds (pre-round state, as DD saw it)

The intervention: one token, two different correct answers

```
DD forces node v because its known neighbours already use q-1 = 3 of the 4 colours.
Exactly one of them, w, holds colour a. So v must take the one colour left: c.

   clean     ... BEGIN_ROUND  v  ->  [c]        w is coloured a, so c is the only option
   corrupt   ... BEGIN_ROUND  v  ->  [a]        recolour w to c; now a is free and DD says a

The ONLY difference between the two prompts is w's colour token.
build_pair asserts (clean != corrupt).sum() == 1, so this is enforced, not assumed.

ROUND 0  (job 50075339): w is a HINT, handed to the model in the prompt.
         Tests whether it applies the elimination rule to GIVEN facts.

CHAINED  (job 50077245): w was DEDUCED by the model in an earlier round, so the
         edited token sits in the REASONING section, not the hints.
         Tests whether it eliminates against facts it produced itself = propagation.

Scored as `corrupt_flipped`: did the prediction at v move to DD's NEW answer a?
Not 'is it still accurate' -- the right answer changes, so memorised output fails.
```

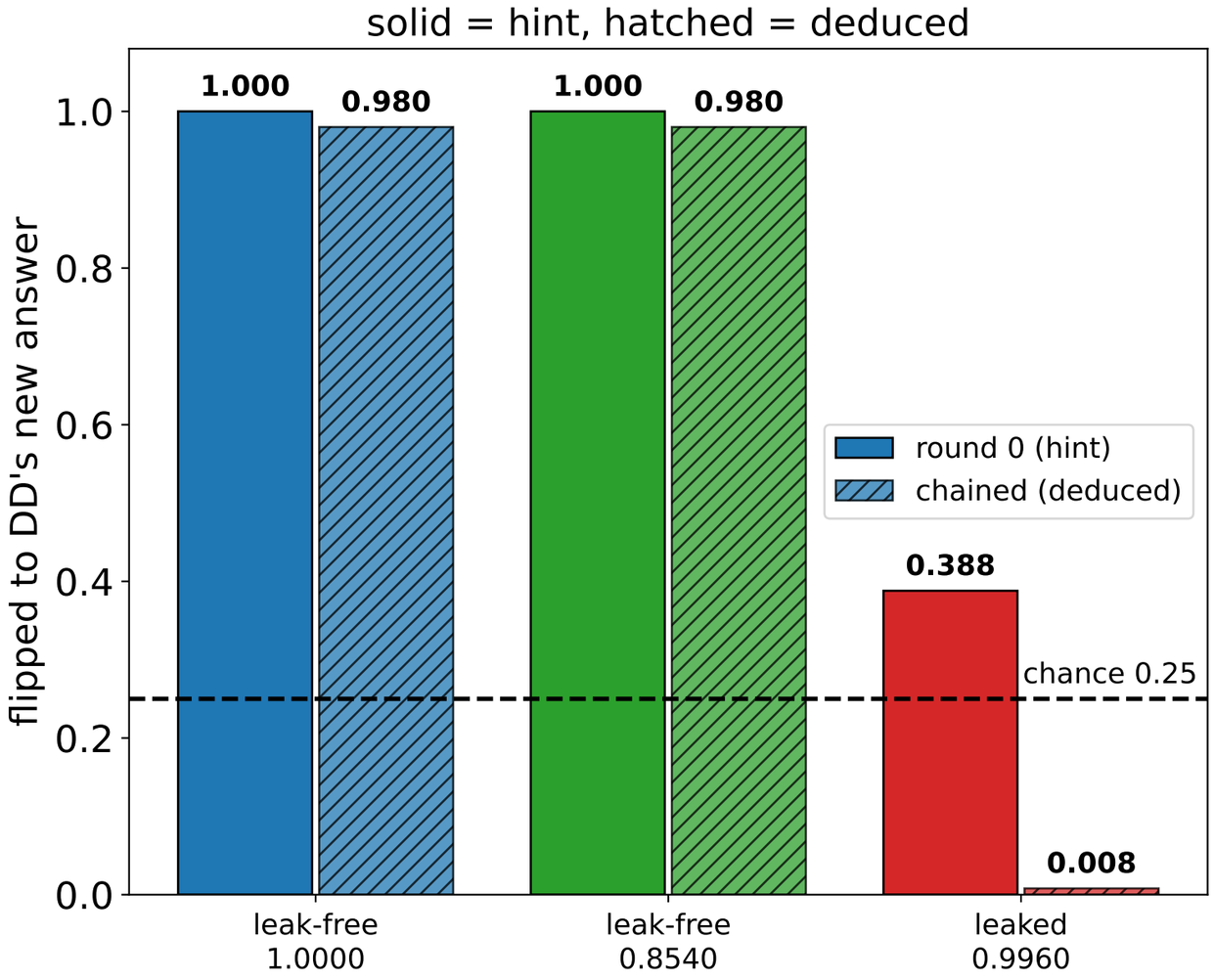
The chained arm's one weak direction: editing an emitted deduction makes the prefix inconsistent, so a model that DETECTS the edit and keeps its own answer scores as 'does not chain'. reverts_to_original separates those cases; see the table.

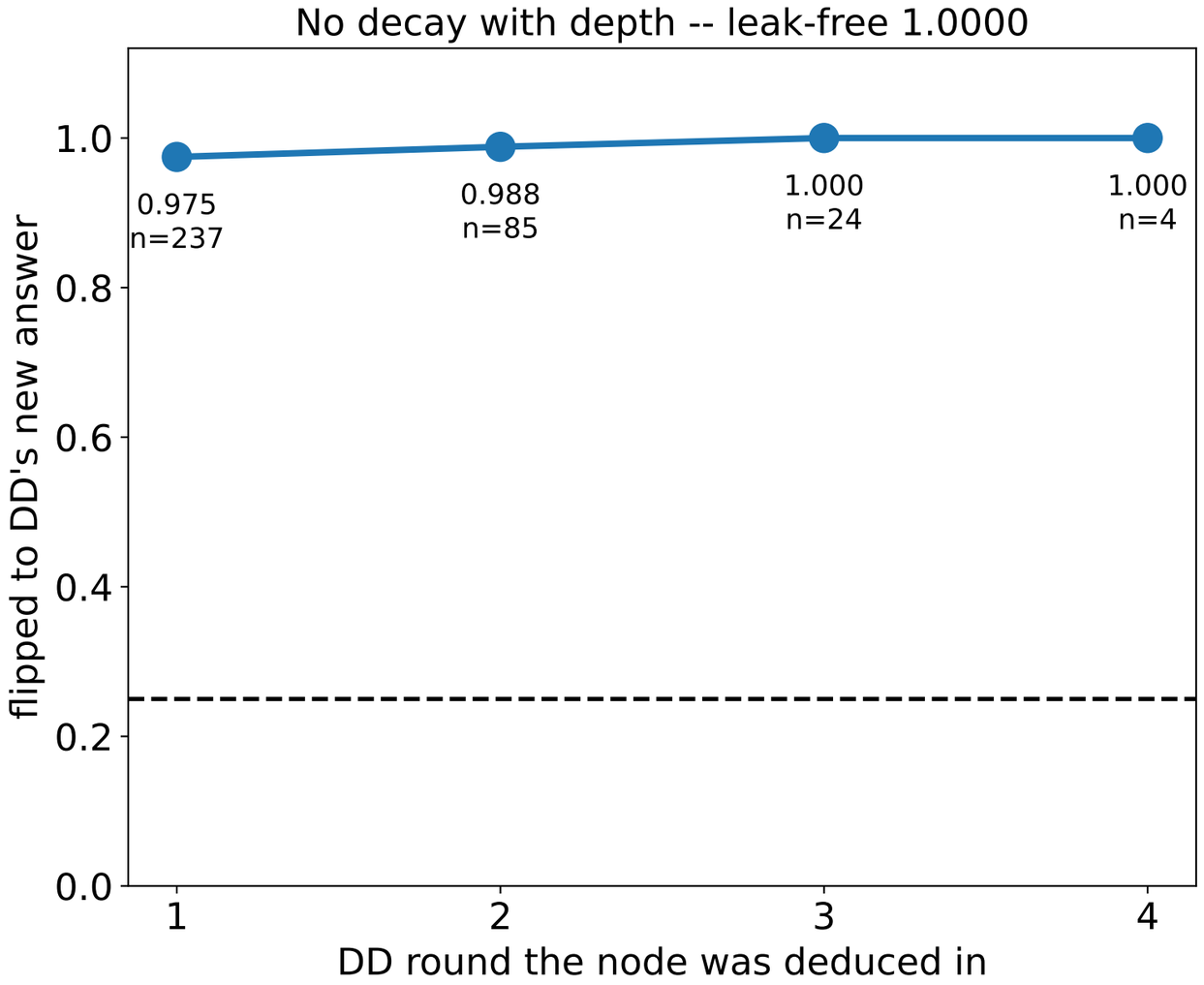
Every causal measurement

arm	test	n	clean	flipped to DD's new answer	reverts to original	mean logit gap clean -> corrupt	verdict
leak-free 1.0000	round 0 (hint)	350	1.000	**1.000**	--	-15.5 -> +15.7	uses the rule
leak-free 1.0000	chained (deduced)	350	1.000	**0.980**	0.020	--	chains
leak-free 0.8540	round 0 (hint)	250	1.000	**1.000**	--	-15.0 -> +14.9	uses the rule
leak-free 0.8540	chained (deduced)	250	1.000	**0.980**	0.008	--	chains
leaked 0.9960	round 0 (hint)	250	1.000	**0.388**	--	-14.5 -> -2.0	near chance
leaked 0.9960	chained (deduced)	250	1.000	**0.008**	0.992	--	near chance

Chance = 0.25. 'clean' is accuracy on the unmodified prompt, so a low 'flipped' with a high 'clean' means the model is right for a reason other than the elimination -- which is exactly the leaked arm's profile.

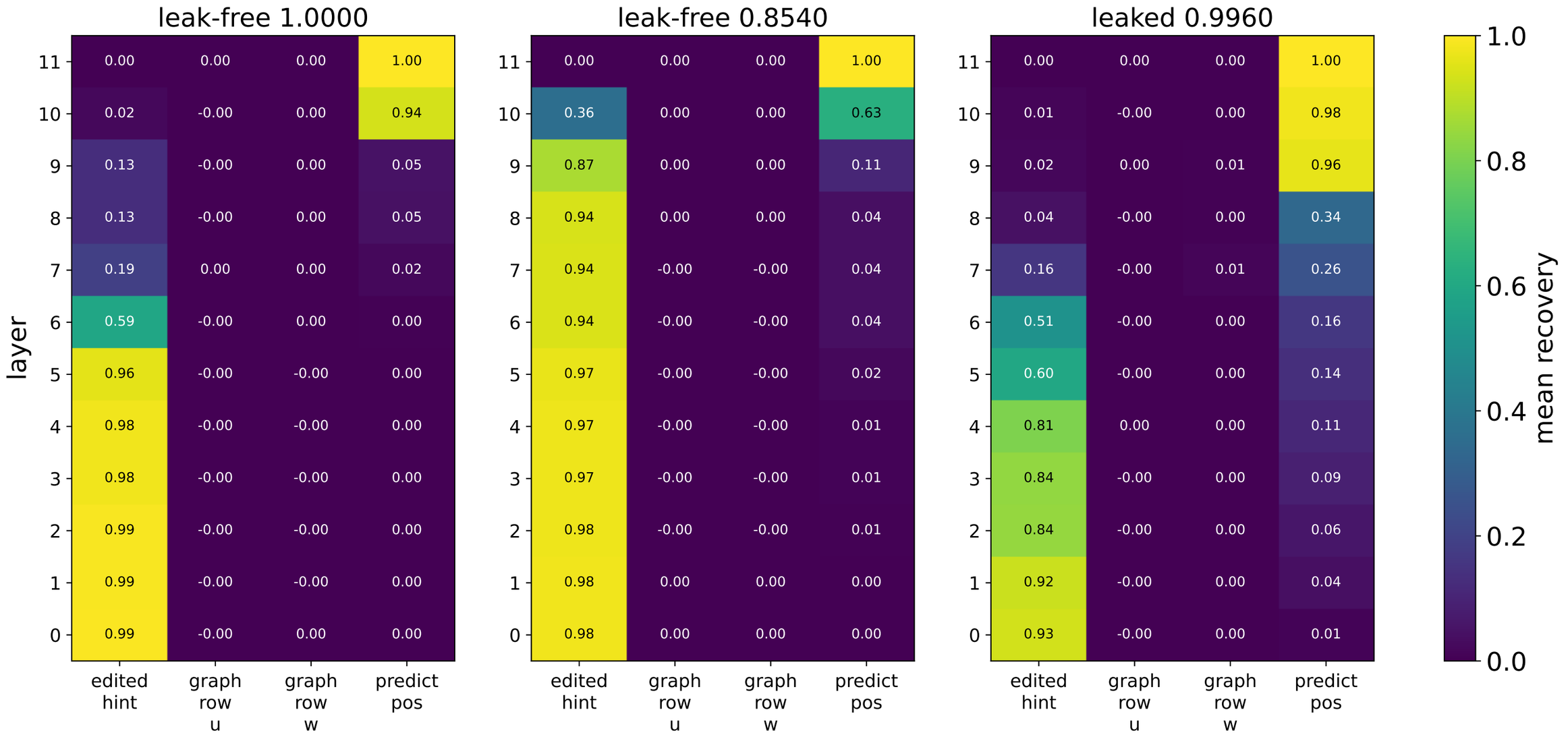
The rule holds on given facts, and on facts it derived itself





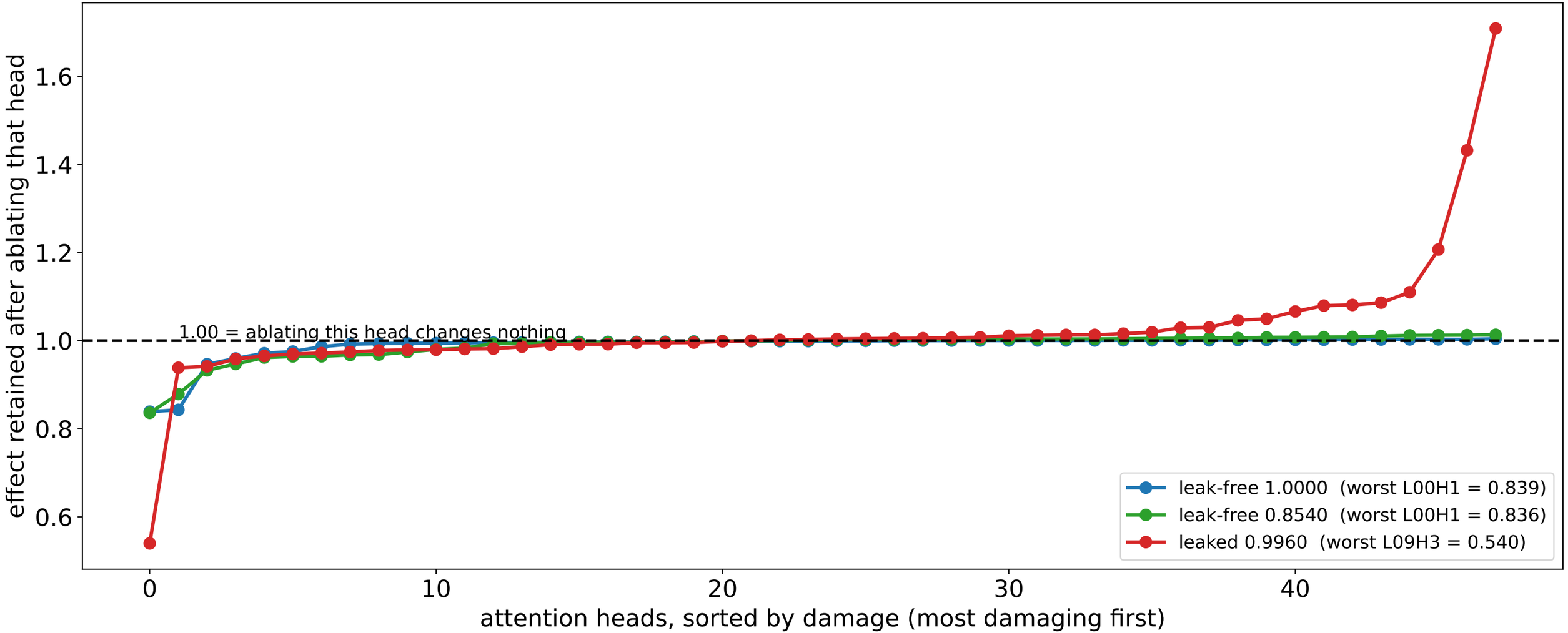
Right panel is the chaining claim: if the model chained one step but not three, round 1 would be high and the later rounds would fall off. It does not happen. The deepest bins are small because deep chained forcings are rare, not filtered.

Where the recoloured fact travels (activation patching)



Recovery = how much of the flip is restored by patching the clean activation at that (layer, site). The fact is held at the EDITED HINT through L5, handed off L6-L10, and is at the PREDICTION POSITION by L10. graph_row_u/graph_row_w are 0.00 at every layer in all arms -- the deduction never routes through the adjacency table rows. L11 at the prediction position is the built-in positive control and must read 1.00.

No single head carries it -- the circuit is distributed



Each point is one of the 48 heads, zeroed one at a time by writing its out_proj weight columns (nn.MultiheadAttention takes the functional fast path, so module hooks on out_proj never fire -- an earlier version of this analysis silently reported 1.000 for all 7200 rows). Removing the worst single head still leaves ~84% of the effect.

The verdict, and what it is not

WHAT IT DOES DO (causal)

* Applies DD's elimination rule: 1.000
  on 350 one-token counterfactuals.

* Chains it off its OWN deductions: 0.980
  No decay from round 1 to round 4.

* Localised: hint -> L5, handoff L6-L10,
  prediction position by L10.

* Distributed across heads, not one head.

WHAT IT DOES NOT DO

* DD's SEARCH SCHEDULE. Events are emitted
  in ascending node index for 100% of rounds.
  A no-deduction hand rule ('smallest index
  whose colour is unknown') scores 0.632 on
  the same max-over-v F1 where the model's
  own next_node features score 0.545.

* Use the adjacency table to do it: the
  graph rows recover 0.000 at every layer.

* The leaked twin does not even do the rule
  (0.388) -- finding 2's 'not running
  DD' was always about THAT model. Shown an
  edited deduction it keeps its own answer
  99.2% of the time: it is reading the
  adjacency-order leak, not the prefix.

Working picture: sweep nodes in index order; at each, eliminate the colours of its known neighbours, including ones it just derived. DD's propagation, not DD's schedule.

Comparison to the previous best answer on this question: finding 2 of this project concluded 'the model is not running DD' from probes, an algorithm-ID test, a counterfactual and an SAE. Its counterfactual is reproduced here as the leaked arm at 0.388. That conclusion stands for the leaked model and is overturned for the leak-free one -- the honest task was never measured before.